MA 2101 : Analysis I

September 8, 2025 Maximum Marks: 20

NOTATIONS

- $\mathbb{R}$, $\mathbb{N}$ denote the set of real numbers and the set of natural numbers respectively.
- sup denotes the supremum or the least upper bound of a set.
- inf denotes the infimum or the greatest lower bound of a set.

Problem 1.

Let E be a non-empty subset of the real numbers. Let $x \in \mathbb{R}$ be such that $x < a$ for all $a \in E$. Then show that $\inf E \geq x$. Prove by an example that the equality can be attained i.e. produce a non-empty subset E of $\mathbb{R}$ such that $x < a$ for all $a \in E$ and $\inf E = x$. [2+3=5]

Problem 2.

Find the supremum and infimum of the set $\{1 - \frac{(-1)^n}{n} : n \in \mathbb{N}\}$. [5]

Problem 3.

(a) Let $I_n = (0, \frac{1}{n})$. Then show that $\cap_n I_n = \emptyset$.

(b) Let $I_n = [0, \frac{1}{n}]$. Then show that $\cap_n I_n = 0$.

[2+2=4]

Problem 4.

(a) Let $I_n = [a_n, b_n]$ where $a_n, b_n \in \mathbb{R}$, $a_n < b_n$ for all $n \in \mathbb{N}$ such that $I_1 \supset I_2 \supset I_3 \supset \ldots$. Then prove that $\cap_n I_n \neq \emptyset$.

(b) Let I_n be a nested sequence of intervals in $\mathbb{R}$. If $\cap_n I_n$ contains more than one point, then show that it contains infinitely many points.

[4+2=6]